

# Vedant Desai

(248) 704 – 4852 | [vedantde@umich.edu](mailto:vedantde@umich.edu) | [github.com/Verdent06](https://github.com/Verdent06) | [linkedin.com/in/vedantde06](https://linkedin.com/in/vedantde06)

## EDUCATION

### University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0

- **Coursework:** Data Structures & Algorithms, Discrete Mathematics

## EXPERIENCE

### CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.
- Eliminated a 27% audio upload failure rate with fault-tolerant RxJS logic that detects expired S3 presigned URLs, regenerates them mid-flight, and negotiates MIME types for WAV files Angular silently rejected.
- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.

### Vylet

May 2026 – Present | [vyletdata.com](https://vyletdata.com)

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Diagnosed a name-collision defect in ownership-verification logic that was incorrectly rejecting valid acquisition targets sharing a name with an unrelated business elsewhere; the fix lifted the pipeline's lead-qualification rate from 79% to 89% with zero change to sourcing volume.

### Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.
- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.

## PROJECTS

### Granular Synthesizer Plugin | C++, JUCE, VST3, AU

[github.com/Verdent06/granular-synth](https://github.com/Verdent06/granular-synth)

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.
- Configured CMake to compile VST3 and AU bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86\_64) for native Apple Silicon and Intel — and audited every `processBlock()` path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before release builds.

### SignalWeaver | Python, FastAPI, React, TypeScript, PostgreSQL

[github.com/Verdent06/SignalWeaver](https://github.com/Verdent06/SignalWeaver)

- Built a React/TypeScript dashboard for composite scores and fundamental/sentiment/macro breakdowns; used it to analyze 90 distinct tickers with scores persisted to Postgres for history views.
- Served composite financial-research scores through async REST endpoints (FastAPI) wrapping fundamentals, MPNet sentiment, and regression logic, instrumented end-to-end at 9.1s p50 / 15.2s p99 across 90 successful runs on 90 tickers.

## TECHNICAL SKILLS

**Languages** ..... Python, TypeScript, C++, SQL, HTML/CSS  
**Frameworks** ..... React, Angular, FastAPI, Flask, RxJS, JUCE  
**Cloud & Infrastructure** ..... AWS (EC2, S3), Docker, Celery, Git